

Linking social relationships at school, loneliness, and academic self-perceptions among primary school children

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ABSTRACT

The present study investigated a theoretical model of the direct and indirect links between children's social relationships at school (i.e., relationships with teachers, peer victimization), loneliness, and academic self-perceptions. Participants were $N = 666$ primary school children (352 girls) aged 7–11 years ($M_{\text{age}} = 8.90$ years, $SD = 1.06$), who completed an online questionnaire. Path analyses indicated that less close teacher–child relationships and higher peer victimization were related to children's loneliness, which, in turn, was associated with less positive academic self-perceptions. Results also revealed some moderating effects of gender. The findings highlight the potentially critical role played by loneliness in determining the association between social relationships at primary school and children's academic self-perceptions.

Academic self-perceptions are defined as a student's self-evaluation of their competencies and abilities in academic domains (Brunner et al., 2009; Harter, 1982; Marsh et al., 2017; Marsh and Martin, 2011). They may be hierarchically organized into domain-general academic self-perceptions (situated at the apex of the hierarchy) and domain-specific academic self-perceptions (Brunner et al., 2009). Positive academic self-perceptions are related to several aspects of school adjustment, including higher academic achievement and more positive attitudes toward school (Green et al., 2012; Marsh and Martin, 2011; Molloy et al., 2011; Weidinger et al., 2018). The present study focused on domain-general academic self-perceptions, which theoretically encompass domain-specific dimensions. We aimed to evaluate a model linking children's social relationships at school, loneliness, and academic self-perceptions. Specifically, we examined conceptual pathways whereby children's perceptions of less close teacher–child relationships and more frequent peer victimization at school are related to higher levels of loneliness (Kingery and Erdley, 2007), which in turn, lead to more negative academic self-perceptions (A. M. Eccles et al., 2023).

Given the academic implications such as those noted above, previous research has aimed at uncovering mechanisms or pathways by which academic self-perceptions develop. The resulting developmental models of academic self-perceptions highlight the important role of social

interactions with significant others (e.g., teachers, peers), particularly during primary school, when children spend more time outside the family context than during the preschool years (Marsh et al., 1998; Verschueren et al., 2012). In this vein, close/supportive teacher–child relationships have been related to children's positive self-perceived academic competencies (Hughes et al., 2012). In contrast, peer victimization, defined as frequent experiences of peer aggression (Kochenderfer and Ladd, 1996), has been associated with poorer academic self-perceptions (Jenkins and Demaray, 2015). To date, little is known about the underlying processes that connect social relationships at school with academic self-perceptions. The present study considers one possibility, specifically loneliness, which is a perceived lack of intimate relationships and/or not belonging to a group (Galanaki and Vassilopoulou, 2007). This may be a particularly relevant factor to consider during the primary school years, as children at this age tend to express an increasing desire to be accepted and included at school (Molinari et al., 2020). This developmental period is also characterized by the increased value placed on friendships and peer (vs. parental) relationships, a greater desire for peer conformity, and increased academic demands (Laursen and Veenstra, 2021; Rubin et al., 2015). Taken together, these converging developmental factors make late childhood and early adolescence important and unique developmental periods in which to

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consider the links between social relationships at school, loneliness, and academic self-perceptions. To the best of our knowledge, no previous study has explored the role of loneliness in the pathways linking social relationships at school and academic self-perceptions at this age period. Finally, our study has applied implications since intervention programs promoting close teacher–child relationships and positive peer relationships at school may also help reduce loneliness and form positive academic self-perceptions (Driscoll and Pianta, 2010; Van Ryzin and Roseth, 2018).

Successful social relationships at school

The present study's theoretical framework was the bioecological model and expectancy-value theory. The bioecological model proposes that individual development may be influenced by children's social relationships in different contexts (Bronfenbrenner and Morris, 2007). During primary school, children face the dual challenges of developing increasingly complex scholastic skills and building age-appropriate relationships (Alford, 2017). In this context, the proximal environment (i. e., microsystem) of social relationships with teachers and peers comprises a critical aspect of children's school life (Den Brok et al., 2004; Furrer and Skinner, 2003; Hughes, 2011), contributing to children's adaptive socio-emotional and academic adjustment (O'Connor et al., 2011; Pianta et al., 2012; Wang et al., 2016). Children who have experienced positive and secure social relationships tend to perceive themselves as more worthy and lovable, whereas children who have experienced negative and insecure relationships tend to view themselves as less worthy and lovable (Thompson, 2008).

Compared to preschoolers, primary school children spend more time at school with teachers and peers, who thus become increasingly influential in children's ability and value perceptions (J. S. Eccles et al., 1983). With this in mind, J. S. Eccles et al. (1983; see also Fan, 2011) developed the expectancy-value theory as a comprehensive conceptual framework for understanding children's social and academic experiences. From this perspective, students' achievement-related choices may be affected by several factors, including the subjective task value, expectations of success, and academic self-perceptions. In line with the bioecological model (Bronfenbrenner and Morris, 2007), the expectancy-value theory also recognizes the relevant role of the attitudes, behaviors, and beliefs of significant socializers (and children's perceptions of these socializers' expectations, beliefs, and attitudes) in affecting children's academic self-perceptions. However, the conceptual mechanisms underlying associations among these constructs remain under-explored. Accordingly, in the present study, we focused on a specific component of this model, explicitly assessing the role of loneliness in “connecting” social relationships at school with children's academic self-perceptions.

High affective quality dyadic teacher–child relationships provide children with emotional support/security and impact positively their socio-emotional functioning and well-being (Lei et al., 2018; Maldonado-Carreño and Votruba-Drzal, 2011). Teachers are also key determinants of children's academic experiences, with children's experiences of positive affect and support in their relationships with teachers linked to their positive academic self-perceptions at different ages (Clem et al., 2021). For example, higher quality teacher–child relationships, characterized by more closeness and less conflict, are positively related to more positive academic self-perceptions in samples of children aged 6–7 (Verschuere et al., 2012) and 10–11 years (McFarland et al., 2016). Similarly, in a sample of older children and adolescents in grades 6–10, Hallinan (2008) reported that students who perceived that their teachers cared about them, respected them, and praised them were more apt to like school. Most recently, Wong et al. (2019) reported that high levels of positive student–teacher relationships were associated with positive academic self-perceptions in math for boys, but not girls, in a large sample of Canadian adolescents aged 15 years.

Peer relationships represent another critical interpersonal dimension of the classroom social context that may influence children's social-emotional and school adjustment (Rubin et al., 2008). As children increase their time spent with peers and build stronger social bonds, their peers' beliefs, behaviors, and values are likely to influence children's mental health, academic self-perceptions, and academic performance (Bukowski and Raufelder, 2019; Fan, 2011). For example, peer victimization in primary school is associated with the development of emotional problems (e.g., loneliness and depression; Ladd and Troop-Gordon, 2003; Rudolph et al., 2011). There is also evidence that higher peer victimization may be related to negative academic self-perceptions (Buhs, 2005), such that victimized children tend to interpret their victimization from peers as indicative of their lack of skills needed for academic success. These findings support Harter (1998), who suggested that difficult social contexts may reinforce children's negative self-evaluation. Thus, we anticipated that peer victimization and difficulties in relationships with teachers might be related to children's negative academic self-perceptions.

Loneliness in the school context

Loneliness is characterized by dissatisfaction with the quantity and/or quality of social relationships—or a discrepancy between one's desired and actual social relationships (Asher and Paquette, 2003; Perlman and Peplau, 1981). Deficits in the quality of friendships and teacher–child relationships are important contributors to the development and maintenance of loneliness (Galanaki and Vassilopoulou, 2007; Vanhalst et al., 2014). For example, when teachers provide emotional support and meet children's needs for security, children may be better equipped to manage loneliness (Galanaki, 2004; Galanaki and Vassilopoulou, 2007). This may be particularly true during the primary school years, when children tend to extend their relationships with significant others outside the family context. In support of this idea, O'Connor et al. (2012) reported that high-quality relationships with teachers protected against the development of long-term internalizing behavior problems in a longitudinal study involving children from pre-kindergarten to grade 5. Of note, internalizing behavior problems were assessed in this study with the *Child Behavior Checklist* (CBCL; Achenbach et al., 1987), which contains items evaluating constructs strongly related to loneliness (e.g., anxiety and negative mood; Sang et al., 2018).

Peer relationships may also influence children's feelings of loneliness, especially during late childhood and early adolescence, when security and acceptance needs are often best met by peers and there is a heightened need for social belonging and acceptance (Asher and Paquette, 2003; LaFontana and Cillessen, 2010; Pedersen et al., 2007). Several studies have found that more positive peer relationships (characterized by, for example, higher friendship quality, heightened levels of peer preference) are related to lower levels of loneliness (Antonopoulou et al., 2019; Liu et al., 2016; Molinari et al., 2020). Conversely, and particularly relevant for the present study, experiences of peer victimization are associated with higher levels of loneliness (Asher and Paquette, 2003; Kochenderfer and Ladd, 1996). For example, Kochenderfer-Ladd and Wardrop (2001) found that children who moved from non-victim to victim trajectories reported increased loneliness from kindergarten to grade 3.

Consequently, loneliness may represent a specific risk factor for children's academic self-perceptions during late childhood and early adolescence, when expanded social relationships play an increasingly important role (A. M. Eccles et al., 2023; Gest et al., 2005). For example, A. M. Eccles et al. (2023) reported that loneliness was related to poorer perceived academic achievement, higher feelings of school dissatisfaction, and greater feelings of school pressure in a Danish sample of early adolescents. According to expectancy-value theory, children's academic self-perceptions are not isolated but develop - and are embedded - within a complex web of social influences (J. S. Eccles et al., 1983; Fan, 2011). However, few researchers have investigated the role of loneliness in

connecting social relationships at school with academic self-perceptions.

Putting it all together: social relationships, loneliness, and academic self-perceptions

The way in which children navigate the school context also influences their feelings and adjustment at school (Wentzel, 2009). Indeed, social connection is a basic human need that directly affects positive emotions such as happiness (Epley and Schroeder, 2014; Sandstrom and Dunn, 2014). In this regard, positive student–student and teacher–student relationships are not only associated with students' lower levels of loneliness (Antonopoulou et al., 2019; Galanaki and Vassilopoulou, 2007), but they have also been shown to predict students' more positive academic self-perceptions (Clem et al., 2021; Verschuere et al., 2012). Moreover, higher levels of loneliness are directly related to less positive academic self-perceptions (A. M. Eccles et al., 2023). Hence, we postulated that children who perceive their relationships with teachers as more satisfying would not only feel less lonely at school, but also perceive their academic abilities more positively. In contrast, children who experienced heightened victimization would tend to experience more loneliness and dissatisfaction with their relationships that, in turn, may be related to less positive academic self-perceptions.

The present study

The present study examined the direct and indirect links between children's perceptions of their social relationships with peers and teachers, loneliness, and academic self-perceptions. Specifically, we tested the role of loneliness in the link between social relationships in the school setting and academic self-perceptions. To the best of our knowledge, no previous study has investigated the specific role played by loneliness in linking social relationships with peers and teachers to academic self-perceptions. The present study aimed at generating insight into the potentially complex and unique underlying processes that may connect close teacher–child relationships and peer victimization with loneliness and academic self-perceptions. We explored these associations in a sample of primary school children, considering the relevance of this developmental stage to the research aims (due to, for example, the increased importance placed on peer relations/friendships, the stronger desire for similarity with peers, and higher academic demands).

Drawing on the above-described theories and empirical findings, we hypothesized a series of direct and indirect effects. First, we expected closer teacher–child relationships to correspond to lower levels of loneliness (Galanaki and Vassilopoulou, 2007) and with more positive academic self-perceptions (Wong et al., 2019). In the same vein, we hypothesized that higher levels of peer victimization would be associated with higher levels of loneliness (Kochenderfer-Ladd and Wardrop, 2001) and with more negative academic self-perceptions (Buhs, 2005; Jenkins and Demaray, 2012, 2015). We also hypothesized that higher levels of loneliness would be associated with more negative academic self-perceptions (A. M. Eccles et al., 2023).

Thus, we expected that supportive and close relationships with teachers would be associated with lower levels of loneliness that, subsequently, would be associated with more positive academic self-perceptions. Conversely, we hypothesized that peer victimization would be associated with higher levels of loneliness that, subsequently, would be associated with more negative academic self-perceptions.

On a more exploratory basis, we sought to examine differences in the associations among these variables as a function of child gender, age, and learning modality. For instance, previous research has produced mixed results for loneliness reported by boys and girls in late childhood and early adolescence, with some studies reporting gender differences (i. e., girls reporting more loneliness than boys) and others not (Coplan et al., 2007; Galanaki, 2004; Rotenberg et al., 2002). Studies have also

found that, in primary school, girls tend to form closer student–teacher relationships than do boys (Koomen and Jellesma, 2015). In light of this evidence, we tentatively postulated that closer teacher–child relationships would be more strongly associated with lower levels of loneliness among girls versus boys. In terms of an age effect, some research has found that, as children move into adolescence, they become increasingly sensitive to peer acceptance and victimization, and negative social peer experiences have an increasingly harmful impact on their mental health, affecting factors such as loneliness (Bukowski and Raufelder, 2019). In this regard, we speculated that the positive relation between peer victimization and loneliness would be stronger among older, compared to younger, children in our sample.

Finally, as an additional novel contribution, the research was conducted in Italy during a specific phase of the COVID-19 pandemic, when schools had recently re-opened after the first national lockdown. While children had returned to school nationally, there continued to be local school closures, depending on levels of viral spread. Accordingly, we explored whether these associations varied as a function of in-person versus online learning modalities. Previous studies have reported that aspects of online learning, such as boredom during online lessons or a lack of social support, relate to negative outcomes, including higher levels of loneliness (e.g., Christ and Gray, 2022; Pozzoli et al., 2022). Young adolescents reported higher levels of loneliness during the lockdown relative to the pre-pandemic era, likely due to their unmet needs for social connection (Sabato et al., 2021). Therefore, we tentatively hypothesized that the negative relation between close teacher–child relationships and loneliness would be stronger among children attending in-person learning modality compared to those engaging with online learning modality.

Method

Participants

Participants were $N = 666$ children ($n = 352$ girls), aged 7–11 years ($M = 8.90$, $SD = 1.06$), attending primary schools across Italy. Approximately 65% of the children's schools were located in southern Italy, 29% in central Italy, and 6% in northern Italy. Of the total sample, 9% ($n = 59$) of the children had an immigrant background, though the child or one or both parents had been born in Italy. During the data collection, $n = 527$ (79%) were experiencing in-person learning, whereas $n = 139$ (21%) were engaging in online learning.

Procedure

The present study comprised part of a larger research project exploring factors related to children's socio-emotional functioning in the school context (Sette et al., 2023). The data were collected through online questionnaires publicized via social media sites (e.g., Facebook) and in primary schools, from December 2020–April 2021, during the COVID-19 pandemic. After parents, children, and school principals (for children involved through schools) provided written consent, children completed self-report assessments in a quiet location at home, via a smartphone/laptop/tablet. Given the young age of some participants, we invited parents to assist their children in case they needed help. Researchers instructed parents to allow their child to complete the questionnaires as autonomously as possible. Participants received no incentive or payment. The study was approved by the Ethics Committee of the Department of Developmental and Social Psychology, Sapienza University of Rome, Italy.

Measures

Child–teacher relationships

Children's perceptions of a close relationship with their main teacher (i.e., characterized by openness, warmth, and security) was measured

using the Italian version of the Student Perception of Affective Relationship with Teacher Scale (SPARTS; Koomen and Jellesma, 2015), which was previously adapted and used with early adolescents in Italy (Marengo et al., 2021). The Closeness subscale is composed of eight items (e.g., “My teacher understands me”) that are rated on a 5-point scale ranging from 1 (*no, that is not true*) to 5 (*yes, that is true*). In the present study, children were asked to complete the measure thinking about their relationship with the main teacher of their class. The subscale displayed good reliability ($\alpha = 0.84$). The scores of all items were averaged to create a total score, with higher scores indicating greater closeness in the teacher–child relationship.

Peer victimization

Children completed the Victimization Scale to investigate their experiences of victimization by peers (Kochenderfer and Ladd, 1996; Ladd and Kochenderfer-Ladd, 2002). Experts translated and then back-translated into Italian the scale. It is comprised of four items (e.g., “When in school, does anyone in your class: Say mean things to you”) that are rated on a 3-point scale ranging from 1 (*never*) to 3 (*a lot*). In the present study, confirmatory factor analysis (CFA) supported a one-factor solution with good model fit, Satorra-Bentler (S-B) $\chi^2(2) = 3.28, p = .19$, CFI = 1.00, RMSEA = 0.03, 90% CI [0.00, 0.09], SRMR = 0.01, and the scale displayed good reliability ($\alpha = 0.71$). The scores of all items were averaged to create a total score, with higher scores indicating higher levels of peer victimization.

General academic self-perceptions

Children’s general academic self-perceptions were assessed using the Self-Description Questionnaire-I (Coplan et al., 2013; Marsh, 1990). Experts translated the questionnaire into Italian and then back-translated it into English. To reduce the time burden on participants, the present study only administered a subset of five items (e.g., “I like all school subjects”), which were rated on a 5-point scale ranging from 1 (*false*) to 5 (*true*). In the present study, CFA supported a one-factor solution with good model fit, S-B $\chi^2(4) = 2.92, p = .57$, CFI = 1.00, RMSEA = 0.00, 90% CI [0.00, 0.05], SRMR = 0.01, and the subscale displayed good reliability ($\alpha = 0.86$). The scores of all items were averaged to create a total score, with higher scores indicating more positive academic self-perceptions.

Loneliness

Child loneliness was measured using the Italian version of the Loneliness and Social Dissatisfaction Questionnaire (Asher and Wheeler, 1985; Caputi et al., 2021). To reduce the time burden on participants, the present study used the shortened “pure” version of the scale (Rotenberg et al., 2004), which includes five items (e.g., “I feel left out of things”). These are rated on a 5-point scale ranging from 1 (*not at all true*) to 5 (*always true*). The Loneliness subscale was adapted and used in a previous study (Sette et al., 2023) to investigate the links between different subtypes of social withdrawal and children’s internalizing difficulties in Italy. The one-factor solution of the scale composed of five items was confirmed using CFA (for details, please see Sette et al., 2023). In the present study, reliability was good ($\alpha = 0.82$). The scores of all items were averaged to create a total score, with higher scores indicating higher levels of loneliness.

Control variables

The path analysis model also included controls for child gender (0 = boys, 1 = girls) and age. We also included the learning modality (0 = *in-person learning*, 1 = *online learning*) as a control variable.

Analytical plan

The descriptive analyses and calculation of correlations among the study variables were conducted using SPSS 27. To assess the conceptual model linking children’s perceptions of their relationships with teachers

and peers at school and their academic self-perceptions via loneliness, we ran a path analysis model in Mplus 8.4 (Muthén and Muthén, 1998–2017) and controlled for gender, age, and learning modality in the association with loneliness and academic self-perceptions. In line with our conceptual model, the path analysis allowed us to test the direct and indirect associations between variables (MacKinnon, 2008). To test the hypothesized indirect effects, we used the bootstrap approach (Mackinnon et al., 2004). In detail, if the 95% confidence interval (CI) obtained via 1000 bootstraps did not include 0, we concluded that the indirect unstandardized effect was statistically significant. However, we should interpret the results with caution due to the cross-sectional nature of the self-reported data.

On an exploratory basis, we also tested the possible moderating effects of child gender, age, and learning modality in conditioning the associations between teacher–child relationships and peer victimization to loneliness and academic self-perceptions and from loneliness to academic self-perceptions. Accordingly, we simultaneously tested in one model all of the two-way interaction terms between gender, age, and learning modality and a close teacher–child relationship and peer victimization (e.g., gender x close teacher–child relationship; gender x peer victimization) in predicting the path model’s mediating and dependent variables. We subsequently analyzed significant interaction terms via simple slope analyses. As suggested by Cohen et al. (2002), we mean-centered all predictor variables prior to calculating interaction terms. We followed Hayes’ (2015) indication by computing the unstandardized index of moderated mediation with 10,000 bias-corrected bootstraps to test formally the presence of moderated mediation. In detail, if the 95% CI of the estimated index of moderated mediation did not include zero, we concluded that the conditional indirect effects were significantly different in the groups of the moderator variable (Hayes, 2015).

We conducted analyses using the full-information maximum-likelihood (FIML) estimation of parameters to handle missing data (Baraldi and Enders, 2010). We assessed the model fit considering a comparative fit index (CFI) > 0.95, root mean square error of approximation (RMSEA) < 0.08, and standardized root mean square residual (SRMR) < 0.06 (Kline, 2010).

Results

Preliminary analyses

Overall, we found that the values for skewness and kurtosis fell within the acceptable range reported by Curran et al. (1996; [2] and [7] for skewness and kurtosis, respectively). The Little (1988) test for missing completely at random (MCAR) was not statistically significant, indicating that the study variables met the strict assumptions of MCAR, $\chi^2(13) = 16.16, p = .24$.

Table 1 displays the descriptive statistics and the inter-correlations between study variables. Of note, close teacher–child relationships were associated with more positive academic self-perceptions and lower levels of peer victimization and loneliness. Older children tended to report less close relationships with teachers. High levels of peer victimization were associated with less positive academic self-perceptions and higher levels of loneliness. Older children reported lower levels of peer victimization. High levels of loneliness were associated with less positive academic self-perceptions. Younger children and those engaging in online learning reported greater feelings of loneliness. Child gender was not significantly associated with any other

Table 1
Descriptive statistics and inter-correlations among study variables.

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6
1. Close teacher–child relationship	3.92	0.83	–					
2. Peer victimization	1.46	0.40	–0.16***	–				
3. Loneliness	1.44	0.63	–0.20***	0.25***	–			
4. Academic self-perceptions	3.73	1.03	0.37***	–0.15***	–0.22***	–		
5. Child age	8.90	1.06	–0.08*	–0.10**	–0.11**	–0.06	–	
6. Gender	–	–	0.04	0.02	–0.01	0.07	–0.02	–
7. Learning modality	–	–	–0.02	–0.01	0.13***	–0.05	–0.01	–0.03

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

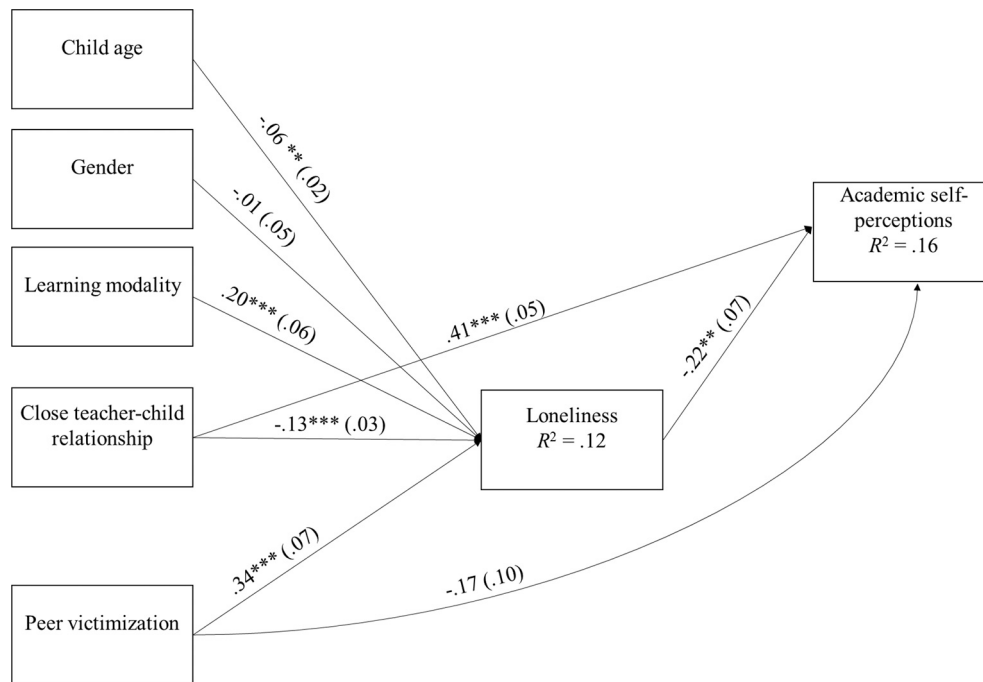


Fig. 1. Links between a close child–teacher relationship, peer victimization, and academic self-perceptions, via loneliness.
Note. We reported the unstandardized values with their respective standard errors in parenthesis. ** $p < .01$, *** $p < .001$.

study variables.

Path model

Fig. 1 presents the results of the path model. We first fitted a model that tested all possible paths, including child age, gender, and learning modality as control variables.¹ Because we tested all possible paths, the initial model had no degrees of freedom and no interpretable fit statistics. In an additional analysis, we sequentially dropped the non-significant paths from the control variables to academic self-perceptions. This final trimmed model is presented in Fig. 1 and showed good model fit, $S-B \chi^2(3) = 5.09$, $p = .16$, $CFI = 0.99$, $RMSEA = 0.03$, 90% CI [0.00, 0.08], $SRMR = 0.02$. The results suggested that close teacher–child relationships related to lower levels of loneliness, which, in turn, were associated with less positive academic self-perceptions. Close teacher–child relationships were also associated with more positive academic self-perceptions. The unstandardized indirect path from close teacher–child relationships to academic self-perceptions via

loneliness was significant and small, $b = 0.03$, $SE = 0.01$, 95% CI [0.01, 0.05], highlighting the role of loneliness in mediating the association between close teacher–child relationships and academic self-perceptions.

The findings also indicated that higher levels of peer victimization were associated with higher levels of loneliness, which, in turn, related to less positive academic self-perceptions. Peer victimization was not significantly associated with academic self-perceptions. The unstandardized indirect path from peer victimization to academic self-perceptions through loneliness was significant and small, $b = -0.07$, $SE = 0.03$, 95% CI [–0.13, –0.02], highlighting the role of loneliness in mediating the association between peer victimization and academic self-perceptions.

For the control variables, the results revealed that younger children felt more lonely than older children. Children engaging in online learning also reported greater loneliness than children experiencing in-person learning. There was no association between gender and loneliness.

Then, further analysis revealed no significant moderating effects of child age or learning modality, although there was a significant moderation effect of child gender in the pathways linking close teacher–child relationships to loneliness ($b = 0.16$, $SE = 0.06$, $p = .01$), and peer victimization to loneliness ($b = 0.28$, $SE = 0.14$, $p = .04$). Because we tested all possible paths simultaneously, this model had no degrees of freedom and no interpretable fit statistics. The model explained 14% of

¹ We also conducted a sensitivity analysis, running the path model with children's immigrant background as a control variable (of note, there was a low percentage of immigrant children in the sample). Here, we present the model omitting that control, as the results were equivalent when this control was included.

the variance of loneliness and 17% of the variance of academic self-perceptions.

In an additional analysis, we dropped the non-significant two-way interaction terms and the non-significant paths from the control variables to academic self-perceptions. The model showed good model fit, $S-B \chi^2(5) = 5.25, p = .38$, CFI = 1.00, RMSEA = 0.01, 90% CI [0.00, 0.06], SRMR = 0.02. The results indicated significant interaction effects between child gender and close teacher–child relationships ($b = 0.16, SE = 0.06, p = .01$) and between child gender and peer victimization ($b = 0.28, SE = 0.14, p = .04$), in relation to loneliness. The results of the follow-up simple effects analyses indicated that closer teacher–child relationships were associated with lower levels of loneliness among boys ($b = -0.21, SE = 0.04, p < .001$), but not girls ($b = -0.06, SE = 0.04, p = .11$; Fig. 2). Additionally, among boys, the unstandardized indirect path from close teacher–child relationships to academic self-perceptions through loneliness was significant and small, $b = 0.05, SE = 0.02, 95\% \text{ CI } [0.01, 0.09]$. Conversely, among girls, the unstandardized indirect path from close teacher–child relationships to academic self-perceptions through loneliness was not significant, $b = 0.01, SE = 0.01, 95\% \text{ CI } [0.00, 0.03]$. Since the 95% CI of the estimated index of moderated mediation (index = 0.03) did not include zero, 95% [0.01, 0.08], we concluded that the mediated path from close teacher–child relationships to academic self-perceptions via loneliness was statistically different between boys and girls.

In contrast, higher levels of peer victimization emerged as strongly related to higher levels of loneliness among girls ($b = 0.46, SE = 0.10, p < .001$), but not boys ($b = 0.18, SE = 0.10, p = .06$; Fig. 3). Among girls, the unstandardized indirect path from peer victimization to academic self-perceptions through loneliness was significant and small, $b = -0.10, SE = 0.04, 95\% \text{ CI } [-0.18, -0.03]$. Among boys, the unstandardized indirect path from peer victimization to academic self-perceptions via loneliness was not significant, $b = -0.04, SE = 0.02, 95\% \text{ CI } [-0.10, 0.00]$. Since the 95% CI of the estimated index of moderated mediation (index = 0.06) did not include zero, 95% [0.01, 0.16], we concluded that the mediated path from peer victimization to academic self-perceptions via loneliness was statistically different between girls and boys. Overall, this model explained 13% of the variance of loneliness and 16% of the variance of academic self-perceptions.

Discussion

Academic self-perceptions are a critical factor in children's school adjustment and academic success (Green et al., 2012; Molloy et al., 2011; Weidinger et al., 2018). In this regard, it is important to identify contributing factors and to elucidate pathways underscoring its

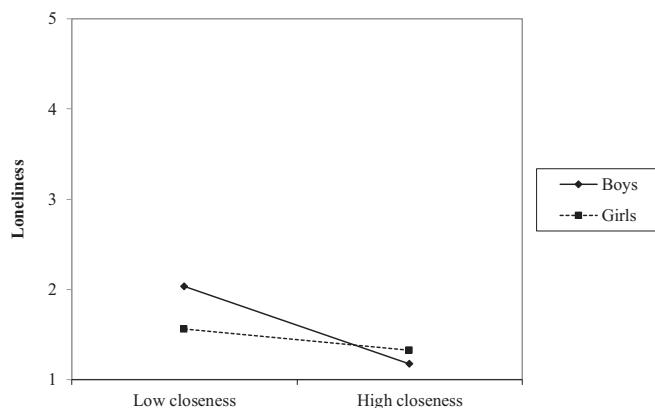


Fig. 2. The moderating role of gender in the association between a close teacher–child relationship and loneliness.

Note. Low closeness (1 SD below the mean), high closeness (1 SD above the mean).

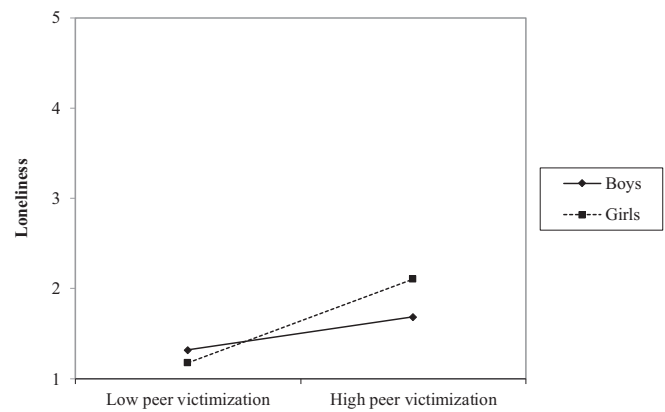


Fig. 3. The moderating role of gender in the association between peer victimization and loneliness.

Note. Low peer victimization (1 SD below the mean), high peer victimization (1 SD above the mean).

development. In line with expectancy-value theory (J. S. Eccles et al., 1983), academic self-perceptions are likely to derive from not only achievement-related experiences, but also social relationships with important others (e.g., teachers, peers). The present study examined the specific role of loneliness in the links between social relationships with peers and teachers and academic self-perceptions during late childhood and early adolescence, when there tends to be high vulnerability to peer victimization and heightened academic demands (Bukowski and Raufelder, 2019).

Overall, the findings indicated that children who perceived their relationships with teachers as close tended to feel less lonely and, in turn, reported more positive academic self-perceptions. In a similar vein, children who reported greater peer victimization felt more loneliness, which in turn led to less positive academic self-perceptions. These results highlight for the first time the potentially important and unique role played by loneliness as a *lynchpin* in the connection between social relationships with peers and teachers and academic self-perceptions among primary school children. These findings also suggest that intervention programs specifically targeting loneliness (e.g., Eccles and Qualter, 2021) may be helpful in interrupting the link from poor social relationships to negative academic self-perceptions.

Social relationships, loneliness, and academic self-perceptions

The primary goal of the present study was to test a conceptual model linking social relationships with peers and teachers and academic self-perceptions among primary school children, via loneliness. In the path model, closer teacher–child relationships related to more positive academic self-perceptions, both directly and indirectly, via lower levels of loneliness. These findings are aligned with theoretical frameworks and research indicating that children who establish closer and more supportive relationships with teachers tend to report lower loneliness (O'Connor et al., 2012) and show greater competence at school (McFarland et al., 2016; Verschueren et al., 2012). In contrast, a lack of supportive relationships with teachers appears to be associated with children's high levels of loneliness and dissatisfaction with the quantity and/or quality of their social relationships (Galanaki and Vassilopoulou, 2007; Vanhalst et al., 2014). In turn, higher levels of loneliness in children represent a specific risk factor for the development of less positive academic self-perceptions (A. M. Eccles et al., 2023). A conceptual pathway may emerge whereby children who perceive their teacher as a secure base for receiving comprehension, help, and support—especially in the face of difficulty—may experience greater levels of support and less loneliness. In turn, these positive feelings at school may help children perceive themselves as more academically able and

competent.

The present findings additionally support the hypothesis that peer victimization contributes to both loneliness and academic self-perceptions. Studies have shown that children who report more frequent exposure to peer aggression also tend to feel more lonely (Kochenderfer-Ladd and Wardrop, 2001; Rotenberg, 2020). In turn, higher levels of loneliness are associated with less positive academic self-perceptions (A. M. Eccles et al., 2023; Gest et al., 2005). In other words, low integration and high difficulty in forming positive peer relationships may be key determinants of children's loneliness (Ribeiro et al., 2019). The present findings are aligned with previous research showing a positive association between peer victimization and internalizing difficulties (e.g., loneliness; Almeida et al., 2021), which, in turn, have been shown to be associated with children's less positive perceptions of their skills, interests, and pleasure taken in school subjects (A. M. Eccles et al., 2023). Taken together, and consistent with the expectancy-value and attachment theories, the present findings support the assertion that both children's social experiences with peers and aspects of the classroom environment contribute to determining children's experiences of loneliness and academic self-perceptions.

Role of gender

On an exploratory basis, we examined whether the links between social relationships, loneliness, and academic self-perceptions varied as a function of child gender. Some significant moderation effects emerged. First, closer teacher-child relationships were associated with less loneliness in boys (but not girls), who, in turn, reported more positive academic self-perceptions. This suggests that there are gender-specific pathways connecting relationships at school with academic self-perceptions. This finding was unexpected, so we only offer some tentative speculations in terms of interpretation. For example, overall, boys tend to form less close relationships with teachers as compared to girls (Koomen and Jellesma, 2015). As such, it is possible that when they create a more positive relationship with teachers, it affords them particular benefits in terms of alleviating feelings of loneliness. Scholars should to replicate these results and further elucidate these postulations in the future research.

In contrast, the present results showed an association between higher levels of peer victimization and higher levels of loneliness in girls (but not boys), who, in turn, reported less positive academic self-perceptions. These findings are aligned with previous studies reporting that, in general, girls are more likely to report internalizing problems (i.e., a broad construct that may include loneliness, anxiety, and negative mood; Sang et al., 2018) following experiences of peer victimization, compared to boys (Crick and Nelson, 2002; Popp et al., 2014). If girls are more sensitive to social acceptance (Rose and Rudolph, 2006), they may also be more likely to internalize, in response to peer difficulties, compared to boys (Graber and Sontag, 2009). In this regard, the impact of negative peer experiences may be more pronounced for girls, than boys.

Limitations and future directions

The present research contributes significantly to the current understanding of the link between social relationships at school, loneliness, and academic self-perceptions. The strengths of the study include the direct assessment of children's perceived quality of relationships with teachers and peers, feelings of loneliness, and academic self-perceptions. The study was also conducted during the school re-opening period after the first national lockdown, and considered children engaging with different learning modalities (i.e., in-person vs. online). We found that children who participated in an online learning modality during the COVID-19 pandemic reported higher levels of loneliness than children who attended school in-person. This is consistent with a growing number of studies indicating that online learning during the COVID-19

pandemic exacerbated feelings of loneliness and other mental health problems among children and adolescents (e.g., Cingel et al., 2022; Crescentini et al., 2020; de Miranda et al., 2020).

Finally, we should also consider some limitations in the interpretation of the results, with an eye toward future research. First, although this is the first study that investigated the role of loneliness in the link between social relationships at school and academic self-perceptions, a notable limitation of the study is that the data were cross-sectional. Therefore, we cannot draw conclusions about developmental processes or establish the temporal ordering of the variables. Thus, it was not possible to evaluate potential bidirectional associations among the study variables over time. Although we explored novel indirect links between social relationships at school and academic self-perceptions via loneliness, researchers should consider alternative causal paths. For instance, child loneliness may impact social relationships with teachers and peers, which, in turn, may affect academic self-perceptions. In support of this notion, Reijntjes et al. (2010) found bidirectional associations between peer victimization and internalizing problems (e.g., anxiety, depression, loneliness) in a meta-analysis of longitudinal studies with samples of children and early adolescents. To address these limitations, scholars can employ longitudinal designs to understand the nature of these inter-associations over time. In particular, it will be important to assess subsequent changes in loneliness and academic self-perceptions as a function of social relationships with teachers and peers. Relatedly, our sample's age range was limited to 7–11 years, and perhaps for this reason, we found limited evidence of age effects. During late childhood and early adolescence, children's vulnerability to peer rejection and victimization tends to increase (Bukowski and Raufelder, 2019). Thus, future research should consider potential differences in the pattern of these associations among different age groups, including older adolescents and emerging adults.

Furthermore, we only considered self-report measures, which may have heightened shared method variance. While teacher reports of student-teacher relationships are commonly employed (Zee et al., 2020), the use of multiple informants and methods (e.g., interviews, observations, questionnaires) may benefit future research (Zuffiano et al., 2018) by capturing different perceptions of social actors in children's school life. Additionally, in the present study, children were allowed to complete their questionnaires at home. It is possible that parents assisted them in this process, even though they were instructed to allow the child to complete the questionnaires as autonomously as possible. It is conceivable that parental presence or assistance may have in some way biased children's responding. As we did not collect information about the number of children who received parental assistance, or where children completed the questionnaires, it is not possible to determine the degree to which this may have impacted our findings. In future studies, children could complete questionnaires at school in order to enforce standardized data collection procedures.

We also evaluated links between social relationships at school and loneliness, which may be considered a specific subset of factors in relation to academic self-perceptions. Scholars should consider a broader evaluation of underlying mechanisms linking social relationships and academic self-perceptions, including the consideration of additional potentially relevant aspects of peer relationships (e.g., friendship quality; Hosan and Hoglund, 2017). It may also be of interest to expand the outcomes beyond academic self-perceptions, to include other related constructs, such as the motivation to learn (Heatly and Votruba-Drzal, 2019). As aforementioned, late childhood and early adolescence are characterized by an increased value placed on friendships and peer (vs. parental) relationships, making this an important developmental period in which to consider social relationships at school and academic self-perceptions.

Implications of the study

The present results have several potential implications for policy and

practice. To begin, the findings underscore the importance of promoting children's social relationships with teachers and peers as important antecedents of school adjustment. In this regard, intervention programs for children, peers, and teachers in the school setting may improve and support children's socio-emotional functioning and well-being. In particular, programs targeting loneliness may be helpful in interrupting the deleterious pathway from poor social relationships to negative academic self-perceptions.

In this direction, previous interventions aimed at fostering more positive teacher–child relationships have been tested with preschool and kindergarten teachers (Driscoll and Pianta, 2010). However, these interventions could be adapted for use with primary school teachers, employing observation and feedback on teaching practices (Sabol and Pianta, 2012). Interventions for victimized or lonely children at school could also be broadly implemented to enhance their inclusion in peer groups during school activities and improve their overall mental health (Kärnä et al., 2011). For instance, teachers may be encouraged to incorporate discussion, group work, role playing, computer games, and short films in their lessons, with the goal of increasing children's empathy toward victims and promoting children's self-efficacy to support victims (Kärnä et al., 2011). Also, structured cooperative school activities aimed at increasing children's opportunities for building positive peer relationships may help to reduce children's victimization and emotional difficulties (Van Ryzin and Roseth, 2018). Finally, intervention programs to promote supportive teacher–child relationships and strong peer relationships (e.g., Driscoll and Pianta, 2010; Van Ryzin and Roseth, 2018) may have a distal/indirect benefit in the form of high youth academic self-perceptions.

Author statement

For this manuscript, we obtained informed consent for experimentation with human subjects. We have observed the privacy rights of human subjects.

Declaration of Competing Interest

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The study was reviewed and approved by the Ethics Committee of the Department of Developmental and Social Psychology, Sapienza University, Rome, Italy. Correspondence concerning this article should be addressed to Stefania Sette, Department of Developmental and Social Psychology, Sapienza University of Rome, via dei Marsi 78, 00185, Rome, Italy. Email: stefania.sette@uniroma1.it, telephone: +39 064991 7921.

Data availability

The dataset for this original article has been uploaded as online supplemental material.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.appdev.2023.101568>.

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